1) Period $= \dfrac{25}{\dfrac{100 \times 10^3}{3600}} = 0.9\,s$

Base excitation frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.9} = \frac{20}{9}\pi$$

$$\approx 6.98 \text{ rad s}^{-1}$$

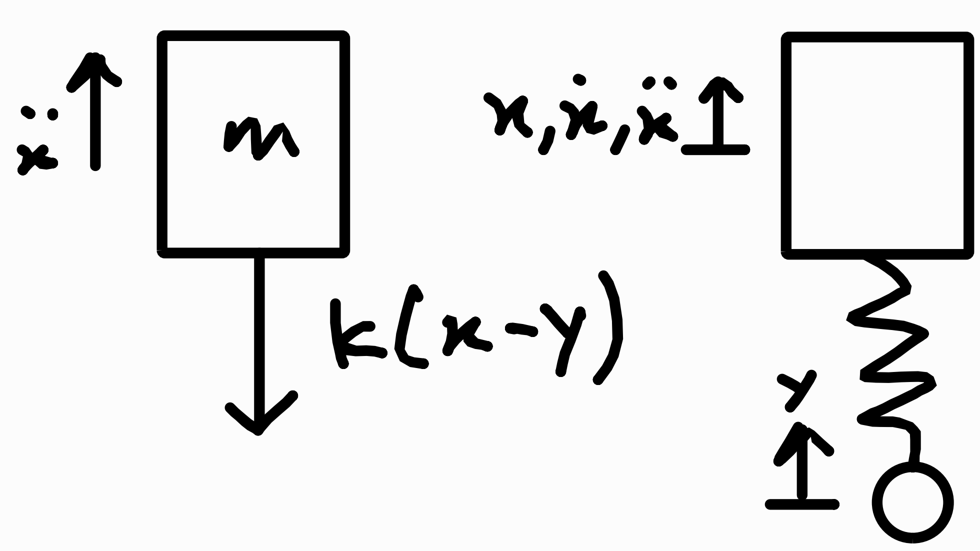
Base excitation amplitude:

$$Y = \frac{15}{2} = 7.5\text{ mm}$$

Base excitation:

$$y(t) = Y\sin\omega t$$

Equation of motion:



$$\sum F_x = m\ddot{x}$$

$$m\ddot{x} = -k(x-y)$$

$$m\ddot{x} = ky - kx$$

$$m\ddot{x} + kx = kY\sin\omega t \quad \because y = Y\sin\omega t$$

2) Maximum vertical excursion:

Steady-state solution is in the form:

$$x(t) = X\sin(\omega t - \phi)$$

Substituting the above into the equation of motion:

$$\dot{x} = \omega X\cos(\omega t - \phi)$$

$$\ddot{x} = -\omega^2 X\sin(\omega t - \phi)$$

$$= -\omega^2 x$$

$$m\ddot{x} + kx = kY\sin\omega t$$

$$-\omega^2 x + kx = kY\sin\omega t$$

$$x(k - m\omega^2) = kY\sin\omega t$$

$$X\sin\omega t\,(k - m\omega^2) = kY\sin\omega t$$

$$X\cancel{\sin\omega t} = \frac{kY\cancel{\sin\omega t}}{k-\omega^2} \quad \because \phi = 0$$

$$X = \frac{kY}{k - m\omega^2}$$

$$= \frac{4\times 25\times 10^3(7.5\times 10^{-3})}{4\times 25\times 10^3 - 1570\left(\frac{20}{9}\pi\right)^2}$$

$$= 0.03194193668$$

$$\approx 0.0319\,m$$

Maximum excursion $= 2X = 2(0.0319)$

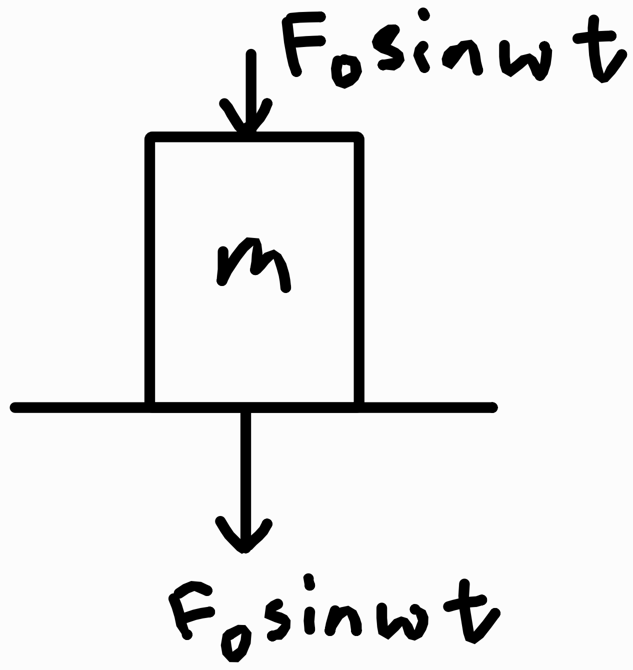
$$= 0.06388387336$$

$$\approx 0.0639\,m$$

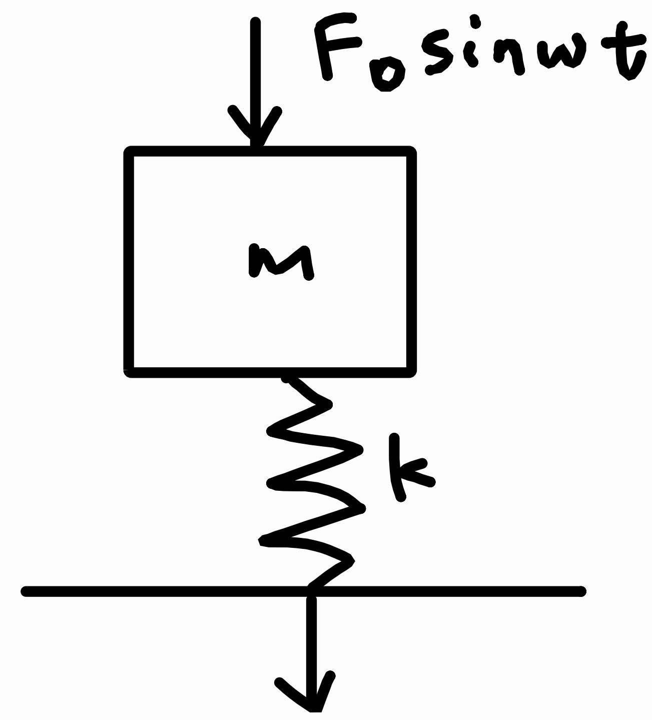
2) $m = 350\,kg$ $\quad f = 800\, rpm$

Any rotating machine has inherent inbalance, so we assume the excitation due to unbalamed forces is of the form $F_0 \sin \omega t$

Directly mounted on the ground



Mounted on rubber pads



$$kx = kX\sin(\omega t - \phi)$$

Amplitude of the dynamic force transmitted is:

$$F_T = kX$$

$$m\ddot{x} + kx = F_0 \sin \omega t$$

$$x = X\sin(\omega t - \phi)$$

$$-m\omega^2 x + kx = F_0 \sin \omega t$$

$$X\cancel{\sin(\omega t - \phi)} = \frac{F_0 \cancel{\sin \omega t}}{k - m\omega^2}$$

$$X = \frac{F_0}{k - m\omega^2}$$

2) $\left|\frac{F_T}{F_0}\right| = \frac{1}{4}$

$$\left|\frac{k\frac{\cancel{F_0}}{k-m\omega^2}}{\cancel{F_0}}\right| = \frac{1}{4}$$

$$\left|\frac{k}{k-m\omega^2}\right| = \frac{1}{4}$$

$$\frac{k}{k-m\omega^2} = \frac{1}{4} \quad \text{or} \quad \frac{k}{k-m\omega^2} = -\frac{1}{4}$$

$4k = k - 350\left(\frac{2\pi}{60}\times 800\right)^2$

$k = -818\,811.6244\ \mathrm{Nm^{-1}}$

$\approx -818.8\ \mathrm{kNm^{-1}}$

(rejected as $k>0$)

$4k = 350\left(\frac{2\pi}{60}\times 800\right)^2 - k$

$k = 491\,286.9746\ \mathrm{Nm^{-1}}$

$\approx 491.3\ \mathrm{kNm^{-1}}$

Static deflection:

$$\Delta_{st} = \frac{mg}{k} = \frac{350\times 9.81}{491.3\times 10^3}$$

$= 6.988786956\ \mathrm{mm}$

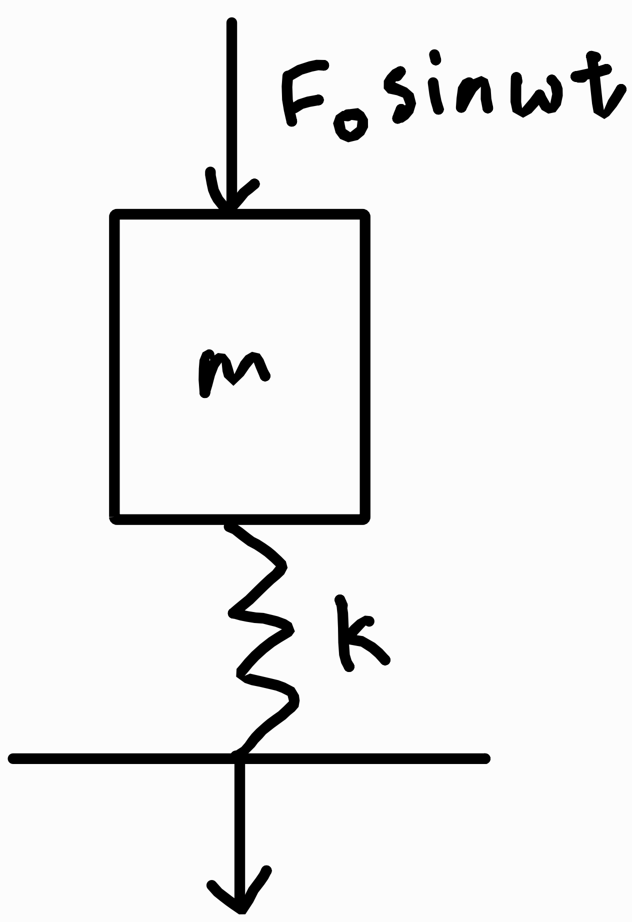
$\approx 6.99\ \mathrm{mm}$

3) $m = 300 \quad \Delta_{st} = 0.005\,m$

$$\Delta_{st} = \frac{mg}{k}$$

$$0.005 = \frac{300 \times 9.81}{k}$$

$$k = 588\,606\,Nm^{-1}$$



$$kx = kX\sin(\omega t - \phi)$$

$$m\ddot{x} + kx = F_0 \sin\omega t$$

$$X = \frac{F_0}{k - m\omega^2}$$

$$\left|\frac{F_T}{F_0}\right| = 0.05$$

$$\left|\frac{k\dfrac{\cancel{F_0}}{k - m\omega^2}}{\cancel{F_0}}\right| = 0.05$$

$$\frac{k}{k - m\omega^2} = \pm 0.05$$

3) $\frac{k}{k-m\omega^2} = 0.05$ or $\frac{k}{k-m\omega^2} = -0.05$

$588600 = 0.05(588600 - 300\omega^2)$

$-15\omega^2 = 559170$

$\omega = 193.0751149i$

$\approx 193.075i \text{ rads}^{-1}$

(rejected since $\omega \in \mathbb{R}$)

$588600 = -0.05(588600 - 300\omega^2)$

$15\omega^2 = 618030$

$\omega = 202.9827579$

$\approx 202.98 \text{ rads}^{-1}$

$\therefore f = \frac{\omega}{2\pi} = \frac{202.98}{2\pi}$

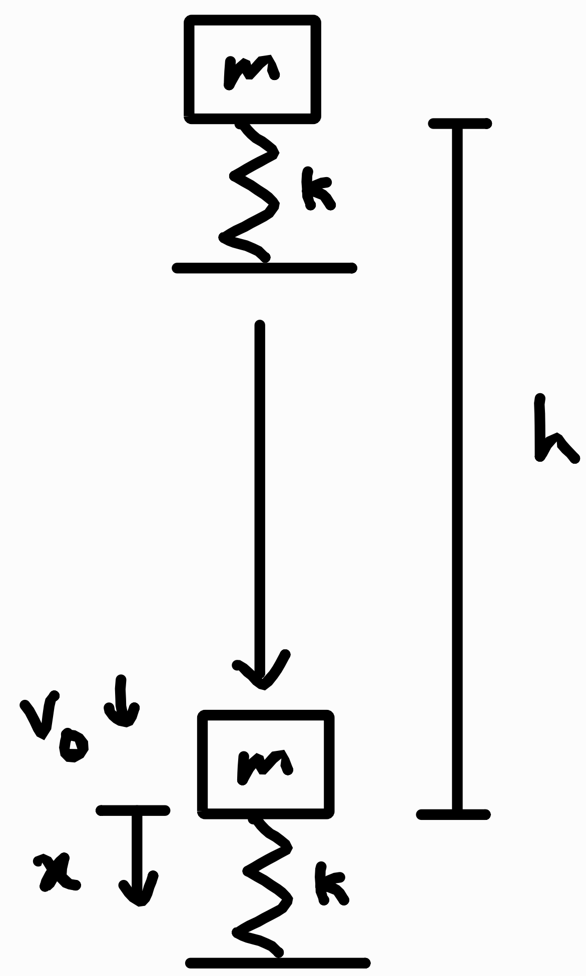
$= 32.30570928$

$\approx 32.3 \text{ Hz}$

4) Velocity of the mass when it contacts the ground is:

$$\frac{1}{2}\cancel{m}v_0^2 = \cancel{m}gh$$

$$v_0 = \sqrt{2gh}$$



The equation of motion:

$$m\ddot{x} + kx = mg$$

The general solution is

$$x(t) = A\cos\omega_n t + B\sin\omega_n t + \frac{mg}{k}$$

When the spring touches the floor:

$$x(0) = 0$$

$$A + \frac{mg}{k} = 0$$

$$A = -\frac{mg}{k}$$

$$\dot{x}(0) = v_0 = \sqrt{2gh}$$

$$\dot{x}(t) = -\omega_n A\sin\omega_n t + \omega_n B\cos\omega_n t$$

$$v_0 = B\omega_n$$

$$B = \frac{v_0}{\omega_n}$$

$\therefore$ The general solution is:

$$x(t) = -\frac{mg}{k}\cos\omega_n t + \frac{\sqrt{2gh}}{\omega_n}\sin\omega_n t + \frac{mg}{k}$$

$$x(t) = \sqrt{\left(\frac{mg}{k}\right)^2 + \left(\frac{v_0}{\omega}\right)^2}\,\sin(\omega_n t + \phi) + \frac{mg}{k}$$

where $\phi = \arctan\left(-\frac{mg\omega_n}{kv_0}\right)$, $v_0 = \sqrt{2gh}$

The force transmitted to the ground is $f_T = kx$.

The reaction force is $R = kx$.

When $R = 0$, the spring loses contact with the ground,

$$\therefore R = kx = 0$$

$$x = 0$$

4) Given $k = 10\,Nm^{-1}$, $m = 1$, $h = 1$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{10}{1}} = \sqrt{10}$$

$$\frac{mg}{k} = \frac{1 \times 9.81}{10} = 0.981$$

$$\phi = \arctan\left(-\frac{1 \times 9.81 \times \sqrt{10}}{10\sqrt{2 \times 9.81 \times 1}}\right) = -0.6109655563$$
$$\approx -0.611\,rad$$

$$v_0 = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 1} = \frac{3\sqrt{218}}{10}$$

$$\sqrt{\left(\frac{mg}{k}\right)^2 + \left(\frac{v_0}{\omega_n}\right)^2} = \sqrt{\left(\frac{1 \times 9.81}{10}\right)^2 + \left(\frac{3\sqrt{218}}{10\sqrt{10}}\right)^2}$$
$$= 1.710076314$$
$$\approx 1.71\,m$$

$$x(t) = 1.71\sin(\sqrt{10}t - 0.611) + 0.981$$

When $x = 0$,

$$0 = 1.71\sin(\sqrt{10}t - 0.611) + 0.981$$

$$\sin(\sqrt{10}t - 0.611) = -0.5736586092$$

$$\sqrt{10}t - 0.611 = -0.611 \text{ or}$$
$$\pi - (-0.611)$$
$$2\pi + (-0.611)$$
$$3\pi - (-0.611) \ldots$$

4) Taking the first value, $t=0$, which is when the spring first touches the ground.

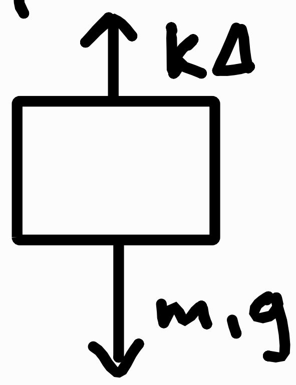
Picking the second solution:

$$\sqrt{10}t - 0.611 = \pi - (-0.611)$$
$$\sqrt{10}t = \pi + 1.221931113$$
$$t = 1.379867373$$
$$\approx 1.380s$$

The second value of $t = 1.380s$ is when the spring first rebounds, hence the spring remains in contact with the ground for 1.380s.

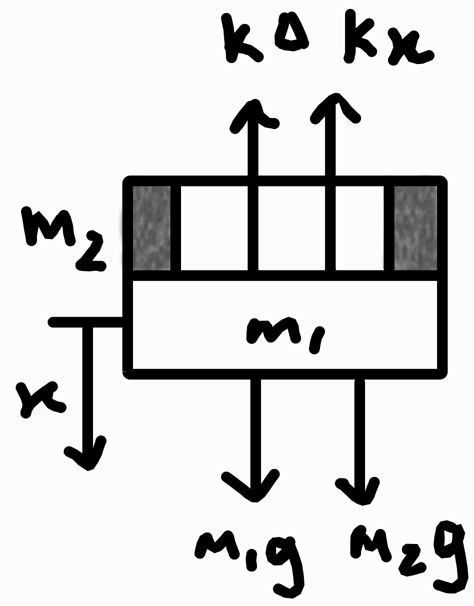
5) Static analysis before impact:

$$k\Delta = m_1 g$$



After impact:

$$(m_1+m_2)\ddot{x} = (m_1+m_2)g - k(\Delta + x)$$



$$(m_1+m_2)\ddot{x} = \cancel{m_1 g} + m_2 g \cancel{- k\Delta} - kx$$

$$\because k\Delta = m_1 g$$

$$(m_1+m_2)\ddot{x} + kx = m_2 g$$

General solution:

$$x = A\sin\omega_n t + B\cos\omega_n t + \frac{m_2 g}{k}$$

$$\dot{x} = A\omega_n\cos\omega_n t - B\omega_n\sin\omega_n t$$

Let v_0 be the velocity of mass m_2 right before impact:

$$\frac{1}{2}\cancel{m_2}v_0^2 = \cancel{m_2}gh$$

$$v_0 = \sqrt{2gh}$$

By conservation of momentum, letting $\dot{x}_0$ be the final velocity of the 2 masses:

$$m_1 \cancel{v_i}^{\,0} + m_2 v_0 = (m_1+m_2)\dot{x}_0$$

$$\dot{x}_0 = \frac{m_2}{m_1+m_2}v_0 = \frac{m_2}{m_1+m_2}\sqrt{2gh}$$

5) When $t = 0$,

$$x(0) = 0, \quad \dot{x}(0) = \dot{x}_0 = \frac{m_2}{m_1+m_2}\sqrt{2gh}$$

$$0 = \cancel{A\sin\omega_n(0)} + B\cos\omega_n(0) + \frac{m_2 g}{k}$$

$$B = -\frac{m_2 g}{k}$$

$$\frac{m_2}{m_1+m_2}\sqrt{2gh} = A\omega_n\cos\omega_n(0) - \cancel{B\sin\omega_n(0)}$$

$$A = \frac{m_2\sqrt{2gh}}{\omega_n(m_1+m_2)}$$

$$x = \frac{m_2\sqrt{2gh}}{\omega_n(m_1+m_2)}\sin\omega_n t - \frac{m_2 g}{k}\cos\omega_n t + \frac{m_2 g}{k}$$

$$= \frac{m_2 g}{k}\left(\frac{k\sqrt{2h}}{\sqrt{g}\,\omega_n(m_1+m_2)}\sin\omega_n t - \cos\omega_n t + 1\right)$$

Substituting $\omega_n = \sqrt{\frac{k}{m_1+m_2}}$,

$$x = \frac{m_2 g}{k}\left(\frac{k\sqrt{2h}}{\sqrt{\frac{gk}{m_1+m_2}}\,(m_1+m_2)}\sin\omega_n t - \cos\omega_n t + 1\right)$$

$$x = m_2\sqrt{\frac{2gh}{k(m_1+m_2)}}\sin\omega_n t + \frac{m_2 g}{k}(1 - \cos\omega_n t)$$